Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/14/2026
PI Name:	Hayman née Anderson, Allison Ph.D.		
Project Title:	Multi-Environment Virtual Training for Long Duration Exploration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	05/19/2021	**End Date:**	09/30/2025
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	8	**No. of Master' Degrees:**	3
No. of Master's Candidates:	11	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	19	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date has changed to 09/30/2025 per A. Beitman/HFBP (Ed., 8/14/25). NOTE: End date has changed to 05/18/2028 per NSSC (Ed., 3/25/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder) Robinson, Stephen Ph.D. (University of California, Davis) Vance, Eric Ph.D. (University of Colorado, Boulder) Szafir, Daniel (University of North Carolina) Miller, Lee (University of California, Davis)		
Grant/Contract No.:	80NSSC21K1140		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of adaptive VR for training in long duration exploration missions (LDEM). These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry decent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (EVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth prior to the start of the mission, and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will experimentally investigate how skills transfer from the training environment to a physical setting, the degree to which the training environment enables skills maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, we will investigate training in an adaptive, complex VR environment as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. The novel contributions of this research are to investigate the use of immersive, adaptive VR environments to train astronauts specifically designed for LDEMs by 1) Incorporating multiple, mission critical tasks for a mission to Mars; 2) Investigating skills transfer, retention, and generalizability; 3) Designing with low volume, mass, and operational overhead; and 4) Investigating the degree to which the proposed training can also provide secondary benefits of broad brain stimulation. Our proposed research is directly aligned with that of the Virtual NASA Specialized Center of Research (VNSCOR), and can be integrated with researchers investigating teams, optimal adaptive control algorithms, motivational strategies, or neural and cognitive decrements with both human and animal studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has wide application in training in immersive environments, particularly for complex tasks. We develop a novel training algorithm to facilitate rapid skills acquisition, while also ensuring broad improvement in performance (rather than unidimensional performance). To our knowledge is absent from the literature, but may improve how we administer training in an automated (rather than supervised) training paradigm. This research will also contribute to our understanding of immersive environments and the degree to which they stimulate the brain. This informs our scientific understanding of neuroplasticity when learning and acquiring new skills. This has applications in rehabilitation, to inform how using these environments may yield better patient outcomes.
Task Progress:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of VR for training in long-duration exploration (LDEM) missions. These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry descent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (sEVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth pre-mission and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission-critical LDEM tasks. We will investigate how skills transfer from the VR training environment to a physical setting, the degree to which the training environment enables performance maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to untrained tasks. As a secondary benefit, training in an adaptive, complex environment could serve as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological responses in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. Specific Aims: This research investigated the following four Specific Aims: Specific Aim 1: Develop a LDEM-specific, multi-environment virtual training simulator We developed a multi-environment training simulator focused on LDEM mission-critical tasks. It is presented using a head-mounted-display (HMD, HTC Vive Pro, and Oculus Quest) and focuses on: 1. Entry, descent, and landing (EDL); 2. Habitat maintenance and repair; and 3. Surface extravehicular activity (EVA) navigation. These existing virtual environments were merged into a single training environment called Trinity that adaptively scales difficulty with user-specific skills acquisition. We developed and evaluated 4 different training algorithms, demonstrating their utility. The training environment provides feedback to the user to increase engagement and motivation. We assessed skills transfer going from the virtual training environment to a physical mock-up test environment (one per environment) in human subject experiments. We quantified skills transfer by evaluating efficiency, errors, and human factors metrics such as workload, situation awareness, and usability. We also investigated highly motivated user populations to understand whether self-led training may be more effective. We found that adaptive difficulty training led to increased skill consolidation and improved performance. Specific Aim 2: Quantify brain activation during training to assess the environment’s efficacy as a countermeasure to neural decrements in spaceflight In Aim 2, we quantified brain activation during training to assess the environment’s potential efficacy in mitigating the cognitive and neural decrements demonstrated in spaceflight. Functional near-infrared spectroscopy (fNIRS) and electroencephalogram (EEG) were used to quantify regional cortical brain activation. We found that the immersive, spaceflight-relevant trainer stimulates regions of the brain more strongly than corollary cognitive tasks without mission relevance. We also investigated whether piloting skills are a precursor to cognitive corollary task performance. Specific Aim 3: Investigate the use of the virtual training environment for performance maintenance This Aim investigated using the multi-environment virtual trainer over longer durations to understand its applicability toward skills and knowledge maintenance, as would be required in transit between the Earth and Mars. This experiment investigated degradation or maintenance across groups with differing levels of training regularity. This experiment is analogous to expected LDEM mission sequences where crews will have access to high-fidelity simulators on Earth in the pre-flight phase, the virtual training environment in the transit phase, and critical task execution, such as upon arrival to the surface of Mars, in the final stage. We found regular use of the virtual trainer led to negative training as subjects reached harder levels, and also that use of the virtual trainer just prior to final task execution in a “just-in-time” fashion mitigated performance deficit. We also investigated the effects of spaceflight and long gaps between training and execution by investigating crew members on the Polaris Dawn mission.

	Specific Aim 4: Demonstrate long-term acceptability, engagement, and skills transfer in the Human Exploration Research Analog (HERA) mission The acceptability of the virtual trainer for use in LDEMs was planned to be evaluated in HERA. We successfully developed an experiment to quantify trainer performance over time, skills transfer to novel tasks not presented in the training environment, sustained engagement, and psychological benefits as a result of repeatedly engaging with the virtual training environment. Due to the cancelation of this grant, this work was not completed.
Bibliography Type:	Description: (Last Updated: 02/18/2026)